

DATA ANALYSIS PROJECT

Customer Shopping Behavior Analysis

Analyzing 3,900 transactions across product categories to uncover spending patterns, customer segments, product preferences, and subscription behavior — guiding strategic business decisions.



Dataset Summary

3,900 Rows × 18 Columns

Transactional data capturing demographics, purchases, and shopping behavior.

Customer Demographics

Age, Gender, Location, Subscription Status.

Purchase Details

Item, Category, Amount, Season, Size, Color.

Shopping Behavior

Discount, Promo Code, Previous Purchases, Frequency, Review Rating, Shipping Type.

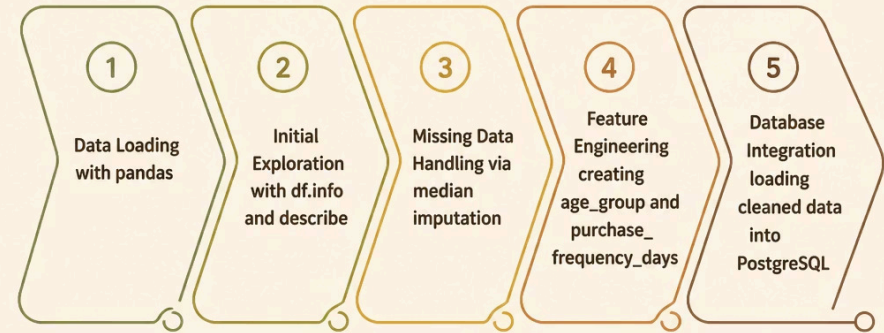
Missing Data Handling

37 null values found in the Review Rating column, imputed using the median rating of each product category.

Exploratory Data Analysis in Python

The data preparation pipeline included loading the dataset with `pandas`, exploring structure via `df.info()` and `.describe()`, and standardizing column names to snake case.

Feature engineering created **age_group** bins and a **purchase_frequency_days** column. Redundant `promo_code_used` was dropped after verifying overlap with `discount_applied`.



Key SQL Insights

\$233K

Total Revenue

Male \$157,890 vs. Female \$75,191

3,116

Loyal Customers

Out of 3,900 total customers

839

High-Spending Discount Users

Discount users above avg spend

27%

Subscribers

1,053 subscribed vs. 2,847 non-subscribed

Top-rated products: Gloves (3.86), Sandals (3.84), Boots (3.82). Most discount-dependent: Hat (50%), Sneakers (49.7%), Coat (49.1%).

Young Adults drive the most revenue (\$62,143).

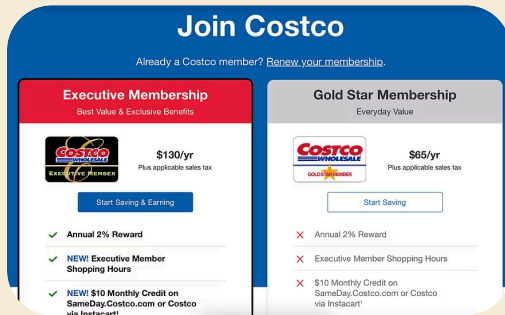
Power BI Dashboard



An interactive dashboard visualizes key metrics across subscription status, gender, category, and shipping type filters.

- 3.9K Customers · \$59.76 Avg Purchase · 3.75 Avg Rating
- Clothing leads revenue (~\$100K), followed by Accessories (~\$60K)
- Young Adults generate the highest revenue across age groups

Business Recommendations



Boost Subscriptions

Promote exclusive benefits to convert the 73% non-subscribers.



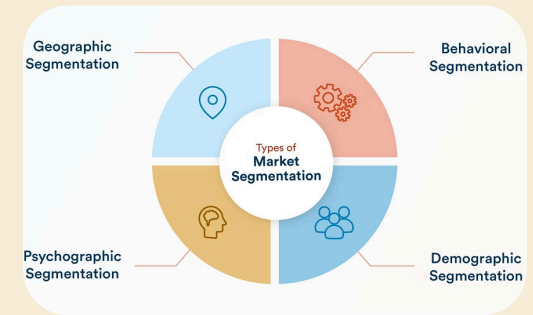
Loyalty Programs

Reward repeat buyers to move them into the Loyal segment.



Review Discount Policy

Balance sales boosts with margin control on high-discount products.



Targeted Marketing

Focus on high-revenue age groups and express-shipping users.